Session 14: Linear Regression 3

December 2, 2024

Cystic fibrosis (CF) is a genetic condition affecting more than 10,600 people in the UK. People with CF experience a build-up of thick sticky mucus in the lungs, digestive system and other organs. Dornase alfa (DNase) is one of the most common cystic fibrosis (CF) treatments. Many clinical trials have found that the use of DNase improves lung function over short periods of time (i.e., less than 2 weeks), however there is little research investigating the longer-term effects of DNase. In this practical, we are interested in the effect of taking DNase for one year on FEV.

We have data on 10,000 individuals in the UK with CF. Out data includes the following information:

- Age at baseline (time 0)
- BMI at baseline
- Sex (male/female)
- FEV at baseline
- Genotype (high/low)
- Lung infection at baseline (yes/no)
- Use of hypertonic saline solution (HSS) at baseline (yes/no)
- Use of DNase (yes/no)
- FEV one year after baseline

where FEV denotes Forced Expiratory Volume (a measure of lung function where higher values indicate better lung function). Genotype is categorised into "High" and "Low". These are genotypes that are either high risk of severe symptoms (high) or low risk of severe symptoms (low).

The baseline time would be when individuals entered the study. Their doctors would then record baseline variables and either prescribe them to take DNase (the treated) or not (the controls). The outcome of interest is FEV measured at a clinical review visit one year later.

This is not a randomised controlled trial. Individuals are not randomised to different treatment groups, rather the decision to initiate treatment with DNase is made by the clinician based on individual characteristics and preferences. Therefore, we expect a number of confounders in this analysis. For this practical, we treat age, BMI, sex, baseline FEV, genotype, HSS use and lung function as confounders.

Question 1: Describe the data Use appropriate graphical tools and summaries to explore the variables in the data. Produce a table of baseline characteristics by treatment group. Summarise the main differences between DNase users and non-users.

Question 2: Univariable analysis Use simple linear regression to estimate the effect of DNase on FEV. Give 95% confidence intervals for the estimated treatment effect and interpret the results.

Question 3: Multivariable analysis Use multivariable linear regression to estimate the effect of DNase on FEV, while adjusting for the listed confounders. Give 95% confidence intervals for the estimated treatment

effects and interpret the results. Comparing the results between the multivariable analysis and the univariable analysis, what can we conclude?

Question 4: Checking assumptions Use appropriate graphical tools to check the validity of assumptions made in the multivariable linear regression model defined in Q3. Are there any influential observations?

Question 5: Further investigations Use multivariable linear regression to investigate the following: (1) does the estimated treatment effect differ for males and females? (2) does the estimated treatment effect depend on FEV at baseline?